PRACTICAL ASSIGNMENT 1.2: REASONING IN KNOWLEDGE BASED EXPERT SYSTEM

NAME: VEDANGI EKNATH PATIL

BATCH: A1

ROLL NO: 19

PRN: 202401040058

Title: SMART PHONE USAGE ADVISOR

Aim:

To develop a Prolog-based Smart Phone Usage Advisor that provides users with information and useful advice about different mobile phone activities based on their battery and mobile data usage.

1. Objectives:

1. To develop a Prolog-based knowledge system that stores information about different smartphone activities.
2. To represent battery usage, mobile data usage, and usage advice for activities such as gaming, video streaming, social media, and online study.
3. To implement an interactive PhoneBot that accepts user input and provides relevant information and recommendations.
4. To understand the use of Prolog concepts such as facts, predicates, rules, backtracking, recursion, and logical querying.

2. Conceptual Details

The Smart Phone Usage Advisor is a simple knowledge-based system developed in Prolog. It provides users with information and advice about

different smartphone activities such as gaming, video streaming, social media, online study, music, video calling, and photography.

The program uses facts and predicates to store information about each activity, including its battery usage, mobile data usage, and useful advice. When the user enters an activity, the system searches the knowledge base and displays the related information.

The phonebot predicate starts the interactive system, while the reply predicate handles different types of user input such as activities, info, list, and stop.

The show_advice predicate uses backtracking to display all advice associated with an activity. If the user enters an unknown activity, the system displays an appropriate error message.

The basic working of the system is:

User Input → Search Knowledge Base → Retrieve Information → Display Advice

3. Conclusion

The Smart Phone Usage Advisor successfully demonstrates the use of Prolog for developing a simple knowledge-based advisory system. The program stores information about smartphone activities, battery consumption, mobile data usage, and useful recommendations.

The interactive PhoneBot allows users to query the knowledge base and receive relevant information. Concepts such as facts, predicates, rules, recursion, backtracking, and logical querying are applied in the implementation.

Overall, the topic shows how Prolog can be used to create a simple decision-support system that provides practical recommendations based on predefined knowledge.

PROLOG :

```
/* ============================================================
   SMART PHONE USAGE ADVISOR
   ============================================================ */
/* ------------------------------------------------------------
   PHONE ACTIVITIES
   ------------------------------------------------------------ */
activity(gaming).
activity(video_streaming).
activity(social_media).
activity(online_study).
activity(music).
activity(video_calling).
activity(photography).
/* ------------------------------------------------------------
   BATTERY USAGE
   ------------------------------------------------------------ */
battery(gaming, high).
battery(video_streaming, high).
battery(social_media, medium).
battery(online_study, medium).
battery(music, low).
battery(video_calling, high).
battery(photography, high).
/* ------------------------------------------------------------
   MOBILE DATA USAGE
   ------------------------------------------------------------ */
data(gaming, high).
data(video_streaming, high).
```

```
data(social_media, medium).
data(online_study, medium).
data(music, medium).
data(video_calling, high).
data(photography, low).
/* ----------------------------------------------------------
   PHONE USAGE ADVICE
   ---------------------------------------------------------- */
advice(gaming, 'Reduce screen brightness.').
advice(gaming, 'Enable battery saver when possible.').

advice(video_streaming, 'Reduce video quality when using mobile data.').
advice(video_streaming, 'Use Wi-Fi when available.').

advice(social_media, 'Turn off unnecessary background activity.').
advice(social_media, 'Reduce unnecessary notifications.').

advice(online_study, 'Use Do Not Disturb mode while studying.').
advice(online_study, 'Keep the screen brightness comfortable.').

advice(music, 'Download songs when possible to save mobile data.').
advice(music, 'Turn off unnecessary background applications.').

advice(video_calling, 'Use Wi-Fi when available.').
advice(video_calling, 'Reduce video quality if the battery is low.').

advice(photography, 'Reduce screen brightness when taking many photos.').
advice(photography, 'Keep enough storage space for new photos.').
/* ----------------------------------------------------------
```

```
   START PHONE ADVISOR
   ------------------------------------------------------- */
phonebot :-
    write('      SMART PHONE USAGE ADVISOR'), nl,
    write('Enter a phone activity to get advice.'), nl,
    write('Type info to see available activities.'), nl,
    write('Type list to see usage details.'), nl,
    write('Type stop to exit.'), nl, nl,
    phone_loop.
/* -------------------------------------------------------
   MAIN LOOP
   ------------------------------------------------------- */
phone_loop :-
    write('PhoneBot :- '),
    read(Input),
    reply(Input).
/* -------------------------------------------------------
   REPLY FOR PHONE ACTIVITY
   ------------------------------------------------------- */
reply(Activity) :-
    activity(Activity),
    battery(Activity, Battery),
    data(Activity, Data),

    write('Activity       : '), write(Activity), nl,
    write('Battery Usage  : '), write(Battery), nl,
    write('Data Usage     : '), write(Data), nl,
    write('Advice         :'), nl,

```

```
    show_advice(Activity),
    nl,

    phone_loop.
/* ----------------------------------------------------------
   SHOW AVAILABLE ACTIVITIES
   ---------------------------------------------------------- */
reply(info) :-
    write('Available phone activities:'), nl,
    write('1. gaming'), nl,
    write('2. video_streaming'), nl,
    write('3. social_media'), nl,
    write('4. online_study'), nl,
    write('5. music'), nl,
    write('6. video_calling'), nl,
    write('7. photography'), nl, nl,

    phone_loop.
/* ----------------------------------------------------------
   SHOW USAGE DETAILS
   ---------------------------------------------------------- */
reply(list) :-
    write('Phone activities and their usage:'), nl,
    activity(Activity),
    battery(Activity, Battery),
    data(Activity, Data),

    write(Activity),
    write(' : Battery: '),
```

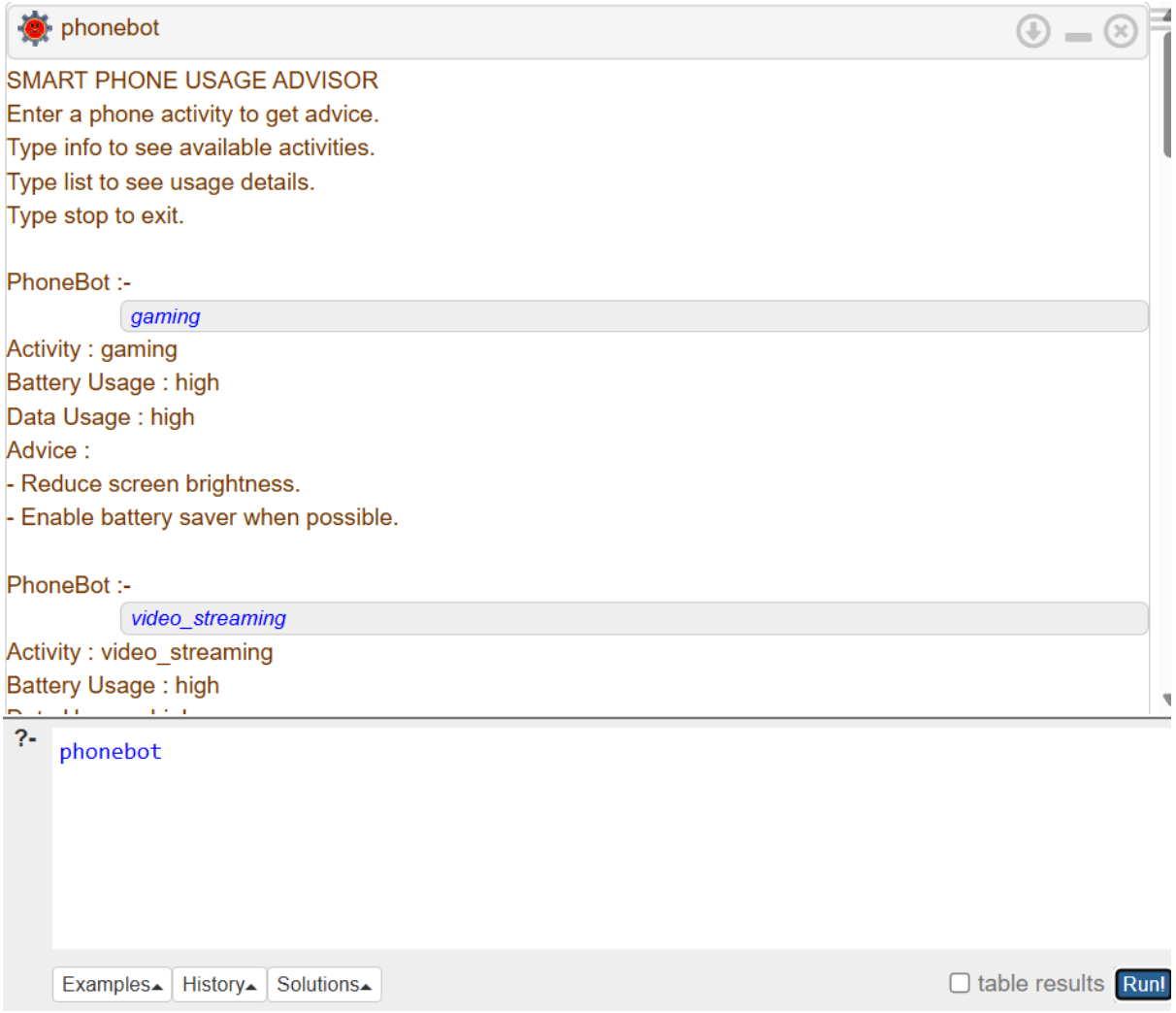
```
    write(Battery),
    write(', Data: '),
    write(Data),
    nl,
    fail.

reply(list) :-
    nl,
    phone_loop.
/* ----------------------------------------------------------
   STOP PHONE ADVISOR
   ---------------------------------------------------------- */
reply(stop) :-
    write('Thank you for using Smart Phone Advisor!'), nl,
    write('Use your phone wisely.'), nl.
/* ----------------------------------------------------------
   UNKNOWN INPUT
   ---------------------------------------------------------- */
reply(Input) :-
    \+ activity(Input),
    Input \= info,
    Input \= list,
    Input \= stop,

    write('Sorry, I do not have advice for '),
    write(Input),
    write('.'), nl,
    write('Please enter a valid phone activity.'), nl, nl,


    phone_loop.
/* ----------------------------------------------------------
   SHOW ALL ADVICE FOR AN ACTIVITY
   ---------------------------------------------------------- */
show_advice(Activity) :-
    advice(Activity, Advice),
    write('- '),
    write(Advice),
    nl,
    fail.
show_advice(_).
```

OUTPUT:


phonebot
SMART PHONE USAGE ADVISOR
Enter a phone activity to get advice.
Type info to see available activities.
Type list to see usage details.
Type stop to exit.
PhoneBot :-
gaming
Activity : gaming
Battery Usage : high
Data Usage : high
Advice :
- Reduce screen brightness.
- Enable battery saver when possible.
PhoneBot :-
video_streaming
Activity : video_streaming
Battery Usage : high
?-
phonebot
Examples
History
Solutions
table results
Run!

Data Usage : high
Advice :
- Reduce video quality when using mobile data.
- Use Wi-Fi when available.

PhoneBot :-
social_media
Activity : social_media
Battery Usage : medium
Data Usage : medium
Advice :
- Turn off unnecessary background activity.
- Reduce unnecessary notifications.

PhoneBot :-
online_study
Activity : online_study
Battery Usage : medium
Data Usage : medium
Advice :
- Use Do Not Disturb mode while studying.

?- phonebot

Examples▲ History▲ Solutions▲ table results Run!

- Keep the screen brightness comfortable.

PhoneBot :-

music

Activity : music
Battery Usage : low
Data Usage : medium
Advice :
- Download songs when possible to save mobile data.
- Turn off unnecessary background applications.

PhoneBot :-

video_calling

Activity : video_calling
Battery Usage : high
Data Usage : high
Advice :
- Use Wi-Fi when available.
- Reduce video quality if the battery is low.

PhoneBot :-

?- phonebot

Examples▲ History▲ Solutions▲ table results Run!

photography

Activity : photography
Battery Usage : high
Data Usage : low
Advice :
- Reduce screen brightness when taking many photos.
- Keep enough storage space for new photos.

PhoneBot :-

info

Available phone activities:
1. gaming
2. video_streaming
3. social_media
4. online_study
5. music
6. video_calling
7. photography

PhoneBot :-

list

Phone activities and their usage:

?- phonebot

Examples▲ History▲ Solutions▲ table results Run!

```
list
Phone activities and their usage:
gaming : Battery: high, Data: high
video_streaming : Battery: high, Data: high
social_media : Battery: medium, Data: medium
online_study : Battery: medium, Data: medium
music : Battery: low, Data: medium
video_calling : Battery: high, Data: high
photography : Battery: high, Data: low

PhoneBot :-
exit
Sorry, I do not have advice for exit.
Please enter a valid phone activity.

PhoneBot :-
web
Sorry, I do not have advice for web.
Please enter a valid phone activity.

PhoneBot :-
```

```
?- phonebot
```

Examples▲ History▲ Solutions▲ table results Run!

```
PhoneBot :-
stop
Thank you for using Smart Phone Advisor!
Use your phone wisely.
true                                                    1
```